

English Presentation Script: Temporary Rental Space Management System of MFU

Estimated Total Time: 20 Minutes

INTRODUCTION & GREETING (~1 Minute)

 **Slide 1: TITLE SLIDE [Paniti]** "Good afternoon, respected committee members. Welcome to our Senior Project. Today, our team is proud to present the 'Temporary Rental Space Management System of MFU'."

 **Slide 2: EXAMINING COMMITTEE [Paniti]** "This our examining committee"

 **Slide 3: PROJECT MEMBERS [Paniti]** "Our team consists of four peoples: I am Paniti Konngoan, acting as the Project Manager and Backend Developer. Joined by Mr. Prathanakorn Piti, our Frontend Developer; Miss Phloiphailin Khampuk, our System Analyst and UX/UI Designer; and Mr. Phongsaphak Duangnet, our Database Administrator and QA. Now, Prathanakorn will start with Chapter 1. Over to you, Prathanakorn."

CHAPTER 1: INTRODUCTION (~3 Minutes)

 **Slide 4: INTRODUCTION [Prathanakorn]** "Thank you, Paniti. Let's begin with Chapter 1: The background and rationale of our project. Currently, the Property Management Division at MFU handles space reservations using a manual, paper-based workflow. This traditional method causes several critical issues: scheduling conflicts, delayed communications, and a heavy administrative burden, especially when coordinating financial transactions with external commercial clients. Our solution is to develop a comprehensive, fully automated web-based platform to digitize this entire workflow."

 **Slide 5: PROJECT OBJECTIVES [Prathanakorn]** "Our project is driven by three main objectives:

1. To develop a fully functional web-based room reservation system for the university.
2. To implement a full-stack system architecture, demonstrating seamless integration between frontend interfaces and backend APIs.
3. To enable secure online bookings for all users and provide administrators with a centralized dashboard to maximize operational efficiency." "Next, Phloiphailin will explain the Scope of Work. Phloiphailin, please."

✓ **Slide 6: SCOPE OF WORK - SYSTEM USERS [Phloiphailin]** "Thank you. Moving on to the Scope of Work. We categorize our system users into three distinct authorization levels: First, **Internal Users** (staff and students). Second, **External Users** (outside companies). And third, **Collaborative Users** (cross-organizational partnerships). The system uses these roles to automatically apply different rental pricing tiers based on university policies."

✓ **Slide 7: SCOPE OF WORK - USER FUNCTIONALITIES [Phloiphailin]** "For the User Functional Scope, individuals can perform real-time availability searches. During the booking process, they can customize their requests by selecting service add-ons—like microphones or extra chairs—and the system will calculate the total cost automatically. Most importantly, users can securely process their payments online, track their live booking status, and submit post-service feedback." "I will now pass the floor to Phongsaphak to discuss the Administrator Controls."

✓ **Slide 8: SCOPE OF WORK - ADMINISTRATOR CONTROLS [Phongsaphak]** "Thank you. For the Administrator Scope, we provide a centralized dashboard. Admins can manage facility details and pricing structures, featuring a bulk import tool using Excel files. The dashboard includes interactive data visualization to monitor revenue and room utilization. Additionally, admins have full control over approval workflows, document management, and promotional broadcasts."

✓ **Slide 9 & 10: INTEGRATIONS & WORKFLOW [Phongsaphak]** "To ensure high security, our system integrates two major third-party services. We use **Google OAuth** for Single Sign-On, which automatically assigns user roles based on their email domains (like @mfu.ac.th). For payments, we chose **Opn Payments** to generate dynamic PromptPay QR Codes, ensuring secure and instant transactions. Overall, our system workflow ensures a seamless transition from user requests to executive approvals, triggering real-time automated alerts at every step."

● CHAPTER 2: LITERATURE REVIEW (~5 Minutes)

✓ **Slide 11: LITERATURE REVIEW - EXISTING VS PROPOSED SYSTEM [Phloiphailin]** "Moving to Chapter 2, Literature Review. When comparing the existing MFU system with our proposed system, the paradigm shift is evident. Currently, external bookings require walk-ins and manual paper routing. We are replacing this with **Direct Online Booking** and a **Fully Automated Web-based Workflow**. Instead of handling cash at the counter, users will experience **Integrated PromptPay QR transactions**. Furthermore, requesting service add-ons will no longer require manual paperwork; it will be an automated selection process with instant cost calculation."

✓ **Slide 12: LITERATURE REVIEW - COMPETITIVE ANALYSIS [Phloiphailin]** "We also conducted a competitive analysis against commercial platforms like Peerspace and Skedda. While those platforms are excellent for general commercial event spaces, they are rigid. Our proposed system is **tailor-made specifically for university**

operations. It features dynamic role-based pricing and a specialized Executive Approval Workflow that standard platforms simply do not accommodate."

✅ **Slide 13: LITERATURE REVIEW - CORE TECHNOLOGIES [Prathanakorn]** "Regarding the Core Technologies chosen for this project: For the **Frontend**, we are utilizing **Vue.js** and **Tailwind CSS**. This combination allows us to build a highly responsive, Single Page Application (SPA) that delivers a smooth user experience across all devices. For the **Backend**, the server logic and API endpoints are handled by **Node.js**, ensuring fast asynchronous processing. And for our **Database**, we rely on **PostgreSQL**, managed via **pgAdmin 4**, to handle complex relational data mapping securely. Now, Paniti will take over to dive deep into our System Analysis and Design."

🟢 CHAPTER 3: ANALYSIS AND DESIGN (~10 Minutes)

(Note: Use this time to show Diagrams and UI Prototypes on the screen)

[Phloiphailin] (Presentation use case diagram)

[Prathanakorn] (Presentation Swim Lane diagram)

[Paniti] "let's explore our **User Interface (UI) Design**." *(Show UI Prototypes or Live Demo)* "We designed the platform adhering strictly to MFU's corporate identity, utilizing deep red and gold tones.

- **For the User Home Page**, we focus on a clean search bar and promotional banners. The room catalog displays clear 'Available' or 'Booked' badges.
- **In the Booking Interface**, users experience a step-by-step form. When they select their user type and add-ons, the right-side panel dynamically calculates the subtotal, applies promo code discounts, and displays the final net price instantly.
- **For the Admin Dashboard**, upon logging in, admins are greeted with data visualization charts summarizing monthly revenue and room utilization rates. The reservation management tab allows admins to quickly approve requests, attach PDF documents, and view post-service feedback from users. This design guarantees an intuitive experience while minimizing human error."

[Paniti] "Moving on to our **Database Design**." *(Show ER Diagram)* "Our ER-Diagram is highly normalized to support scalability. For instance, the rooms and room_pricing tables are separated using a One-to-Many relationship. This allows a single room to hold multiple dynamic prices based on user roles and durations (Half-day or Full-day). Another crucial component is the admin_activity_logs table. To ensure security and accountability, this table records every single action an admin takes, storing the IP address and the exact JSON data before and after the modification."

● CHAPTER 4: PROJECT PROGRESS AND CONCLUSION (~2 Minutes)

✓ **Slide 14 & 15: TIMELINE & RESOURCES [Paniti]** "Bringing us to Chapter 4: Project Progress and Future Work. Looking at our timeline, during this past term (January to May), our team has successfully completed Phase 1 and Phase 2. This includes comprehensive requirement analysis, system architecture design, database modeling, and the development of the functional UI prototype you just saw. (15) For our resources, moving forward, we will utilize actual room inventory datasets provided by the Property Management Division to populate our database."

✓ **Slide 16 & 17: EXPECTED RESULTS & CONCLUSION [Paniti]** "For our next phase in Senior Project 2 (August to December), we will focus entirely on full-stack implementation. We will integrate the Google OAuth and Open Payments APIs, followed by rigorous software testing to resolve bugs and guarantee system stability.

Ultimately, the successful deployment of this web-based system will completely modernize MFU's room reservation process. It will eradicate manual paperwork, optimize facility utilization, and significantly enhance the university's commercial revenue generation capabilities.

✓ **Slide 18: Q&A** "Thank you very much for your time and attention. We would now like to open the floor to the committee for any questions, comments, or technical feedback. Thank you."